

■ Python Practice Problems

Level: **HARD**

35

Problems

Topics covered: **Lists • Tuples • Dictionaries • Functions • Loops • Conditions**

Practice • Learn • Master Python

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Problem 66

Longest Palindromic Substring

Tags: String | List | Loop

Given a string **s**, return the longest palindromic substring. If there are multiple answers of the same length, return the first one found.

Example 1:

Input: `s="babad"`

Output: `'bab'`

Explanation: 'aba' is also valid

Example 2:

Input: `s="cbbd"`

Output: `'bb'`

Example 3:

Input: `s="a"`

Output: `'a'`

Problem 67

LRU Cache

Tags: Dict | Function | Class

Implement an LRU (Least Recently Used) cache class with methods **get(key)** and **put(key, value)**. The cache should evict the least recently used key when capacity is exceeded. **get** returns -1 if key not found.

Example 1:

Input: `LRU(2); put(1,1); put(2,2); get(1)=1; put(3,3); get(2)`

Output: `-1`

Explanation: Key 2 was evicted

Example 2:

Input: `LRU(1); put(1,1); put(2,2); get(1)`

Output: `-1`

Example 3:

Input: `LRU(2); put(1,1); get(1)`

Output: `1`

Problem 68

Word Ladder Length

Tags: Dict | List | Loop | Function

Given **beginWord**, **endWord**, and a **wordList**, return the length of the shortest transformation sequence from beginWord to endWord, where each step changes exactly one letter. Return 0 if no path exists.

Example 1:

```
Input: begin="hit",end="cog",wordList=["hot","dot","dog","lot","log","cog"]  
Output: 5
```

Example 2:

```
Input: begin="hit",end="cog",wordList=["hot","dot","dog","lot","log"]  
Output: 0
```

Example 3:

```
Input: begin="a",end="c",wordList=["a","b","c"]  
Output: 2
```

Problem 69

Merge K Sorted Lists

Tags: List | Loop | Function

Given a list of k sorted lists **lists**, merge all sorted lists into one sorted list and return it.

Example 1:

```
Input: lists=[[1,4,5],[1,3,4],[2,6]]  
Output: [1,1,2,3,4,4,5,6]
```

Example 2:

```
Input: lists=[]  
Output: []
```

Example 3:

```
Input: lists=[[[]]]  
Output: []
```

Problem 70

Minimum Window Substring

Tags: String | Dict | Loop

Given strings **s** and **t**, return the minimum window substring of **s** such that every character in **t** is included. Return "" if no such window exists.

Example 1:

Input: s="ADOBECODEBANC", t="ABC"

Output: 'BANC'

Example 2:

Input: s="a", t="a"

Output: 'a'

Example 3:

Input: s="a", t="aa"

Output: ''

Problem 71

Serialize Nested Dict

Tags: Dict | Recursion | Function | String

Write a function **serialize(d)** that converts a nested dictionary into a flat dictionary by joining keys with a dot. E.g. {'a':{'b':1}} becomes {'a.b':1}.

Example 1:

Input: d={'a':{'b':{'c':1}}, 'd':2}

Output: {'a.b.c':1, 'd':2}

Example 2:

Input: d={'x':{'y':10}, 'z':{'w':20}}

Output: {'x.y':10, 'z.w':20}

Example 3:

Input: d={'key':'value'}

Output: {'key':'value'}

Problem 72

Find All Permutations

Tags: List | Recursion | Function

Given a list of distinct integers **nums**, return all possible permutations as a list of lists.

Example 1:

Input: `nums=[1, 2, 3]`

Output: `[[1, 2, 3], [1, 3, 2], [2, 1, 3], [2, 3, 1], [3, 1, 2], [3, 2, 1]]`

Example 2:

Input: `nums=[1]`

Output: `[[1]]`

Example 3:

Input: `nums=[1, 2]`

Output: `[[1, 2], [2, 1]]`

Problem 73

Median of Two Sorted Lists

Tags: List | Loop | Function

Given two sorted lists **nums1** and **nums2**, return the median of the two sorted lists combined. The overall run time complexity should be $O(\log(m+n))$.

Example 1:

Input: `nums1=[1, 3], nums2=[2]`

Output: `2.0`

Example 2:

Input: `nums1=[1, 2], nums2=[3, 4]`

Output: `2.5`

Example 3:

Input: `nums1=[], nums2=[1]`

Output: `1.0`

Problem 74

Reconstruct Itinerary

Tags: Dict | List | Recursion | Function

Given a list of airline tickets **tickets** as (from, to) pairs, reconstruct the itinerary in order starting from 'JFK'. Use all tickets exactly once. Return the lexicographically smallest valid itinerary.

Example 1:

```
Input: tickets=[['MUC', 'LHR'], ['JFK', 'MUC'], ['SFO', 'SJC'], ['LHR', 'SFO']]
Output: ['JFK', 'MUC', 'LHR', 'SFO', 'SJC']
```

Example 2:

```
Input:
tickets=[['JFK', 'SFO'], ['JFK', 'ATL'], ['SFO', 'ATL'], ['ATL', 'JFK'], ['ATL', 'SFO']]
Output: ['JFK', 'ATL', 'JFK', 'SFO', 'ATL', 'SFO']
```

Example 3:

```
Input: tickets=[['JFK', 'KUL'], ['JFK', 'NRT'], ['NRT', 'JFK']]
Output: ['JFK', 'NRT', 'JFK', 'KUL']
```

Problem 75

Alien Dictionary

Tags: Dict | List | Function

Given a sorted list of words in an alien language **words**, derive the order of the alien alphabet. Return a string of characters in the correct order. Return "" if invalid.

Example 1:

```
Input: words=["wrt", "wrf", "er", "ett", "rftt"]
Output: 'wertf'
```

Example 2:

```
Input: words=["z", "x"]
Output: 'zx'
```

Example 3:

```
Input: words=["z", "x", "z"]
Output: ''
```

Explanation: Invalid (cycle)

Problem 76

Largest Rectangle in Histogram

Tags: List | Loop | Function

Given an array of integers **heights** representing histogram bar heights (width=1 each), return the area of the largest rectangle.

Example 1:

Input: heights=[2,1,5,6,2,3]

Output: 10

Example 2:

Input: heights=[2,4]

Output: 4

Example 3:

Input: heights=[1]

Output: 1

Problem 77

N-Queens

Tags: List | Recursion | Function

Given an integer **n**, return all distinct solutions to the n-queens puzzle. Each solution represents a board configuration where 'Q' is a queen and '.' is empty.

Example 1:

Input: n=4

Output: [[".Q...", "...Q", "Q...", "...Q."], ["...Q.", "Q...", "...Q", ".Q..."]]

Example 2:

Input: n=1

Output: [["Q"]]

Example 3:

Input: n=2

Output: []

Problem 78

Regex Matching

Tags: String | Recursion | Dict | Function

Implement regex matching with '.' (matches any single char) and '*' (matches zero or more of the preceding element). The match must cover the entire string **s**.

Example 1:

Input: `s="aa", p="a"`

Output: `False`

Example 2:

Input: `s="aa", p="a*"`

Output: `True`

Example 3:

Input: `s="ab", p=".*"`

Output: `True`

Problem 79

Trapping Rain Water

Tags: List | Loop | Function

Given a list **height** of non-negative integers representing elevation map bar widths, compute how much water can be trapped after raining.

Example 1:

Input: `height=[0,1,0,2,1,0,1,3,2,1,2,1]`

Output: `6`

Example 2:

Input: `height=[4,2,0,3,2,5]`

Output: `9`

Example 3:

Input: `height=[1,0,1]`

Output: `1`

Problem 80

Expression Evaluator

Tags: String | List | Dict | Function

Write a function **evaluate(expr)** that evaluates a basic arithmetic expression string containing +, -, *, / (integer division), and parentheses. Assume no spaces.

Example 1:

Input: `expr="3+2*2"`

Output: 7

Example 2:

Input: `expr="(2+3)*4"`

Output: 20

Example 3:

Input: `expr="10/(2+3)"`

Output: 2

Problem 81

Interleave Lists

Tags: List | Loop | Function

Given k lists of various lengths, interleave them element by element (round-robin). Shorter lists are skipped when exhausted.

Example 1:

Input: `lists=[[1,2,3],[4,5],[6,7,8,9]]`

Output: [1,4,6,2,5,7,3,8,9]

Example 2:

Input: `lists=[[1],[2],[3]]`

Output: [1,2,3]

Example 3:

Input: `lists=[[1,2,3]]`

Output: [1,2,3]

Problem 82

Decode Ways

Tags: String | Dict | Loop | Function

A message encoded with A=1, B=2, ..., Z=26. Given a string **s** of digits, return the number of ways to decode it.

Example 1:

Input: s="12"

Output: 2

Explanation: 'AB' (1,2) or 'L' (12)

Example 2:

Input: s="226"

Output: 3

Example 3:

Input: s="06"

Output: 0

Problem 83

Minimum Cost Path

Tags: List | Dict | Loop | Function

Given a 2D grid **grid** of non-negative integers, find the path from top-left to bottom-right (moving only right or down) with minimum total cost.

Example 1:

Input: grid=[[1,3,1],[1,5,1],[4,2,1]]

Output: 7

Explanation: 1→3→1→1→1

Example 2:

Input: grid=[[1,2],[1,1]]

Output: 3

Example 3:

Input: grid=[[5]]

Output: 5

Problem 84

Longest Increasing Subsequence

Tags: List | Loop | Function

Given an integer list **nums**, return the length of the longest strictly increasing subsequence.

Example 1:

Input: `nums=[10,9,2,5,3,7,101,18]`

Output: 4

Explanation: [2,3,7,101]

Example 2:

Input: `nums=[0,1,0,3,2,3]`

Output: 4

Example 3:

Input: `nums=[7,7,7,7,7]`

Output: 1

Problem 85

Course Schedule

Tags: Dict | List | Loop | Function

Given **numCourses** and a list of **prerequisites** pairs [a, b] (must take b before a), return **True** if you can finish all courses (no cycle), else **False**.

Example 1:

Input: `numCourses=2, prerequisites=[[1,0]]`

Output: True

Example 2:

Input: `numCourses=2, prerequisites=[[1,0],[0,1]]`

Output: False

Example 3:

Input: `numCourses=1, prerequisites=[]`

Output: True

Problem 86

Scramble String

Tags: String | Dict | Recursion | Function

We can scramble a string by splitting it into two non-empty parts and optionally swapping them, recursively. Given **s1** and **s2**, return **True** if **s2** is a scramble of **s1**.

Example 1:

Input: s1="great", s2="rgeat"

Output: True

Example 2:

Input: s1="abcde", s2="caebd"

Output: False

Example 3:

Input: s1="a", s2="a"

Output: True

Problem 87

Skyline Problem

Tags: List | Loop | Function

Given a list of buildings **buildings** as [left, right, height], return the skyline as a list of [x, height] critical points.

Example 1:

Input: buildings=[[2,9,10],[3,7,15],[5,12,12],[15,20,10],[19,24,8]]

Output: [[2,10],[3,15],[7,12],[12,0],[15,10],[20,8],[24,0]]

Example 2:

Input: buildings=[[0,2,3],[2,5,3]]

Output: [[0,3],[5,0]]

Example 3:

Input: buildings=[[1,5,11]]

Output: [[1,11],[5,0]]

Problem 88

Count Inversions

Tags: List | Loop | Function

Given a list **nums**, count the number of inversions — pairs (i, j) where $i < j$ but $\text{nums}[i] > \text{nums}[j]$. Use a merge-sort approach for $O(n \log n)$.

Example 1:

Input: `nums=[3, 1, 2]`

Output: 2

Explanation: (3,1) and (3,2)

Example 2:

Input: `nums=[1, 2, 3, 4]`

Output: 0

Example 3:

Input: `nums=[4, 3, 2, 1]`

Output: 6

Problem 89

Partition Equal Subset Sum

Tags: List | Dict | Function

Given a non-empty integer list **nums**, determine if it can be partitioned into two subsets with equal sums.

Example 1:

Input: `nums=[1, 5, 11, 5]`

Output: True

Explanation: [1,5,5] and [11]

Example 2:

Input: `nums=[1, 2, 3, 5]`

Output: False

Example 3:

Input: `nums=[2, 2, 1, 1]`

Output: True

Problem 90

Shortest Path in Grid

Tags: List | Dict | Loop | Function

Given a binary grid **grid** (0=open, 1=blocked), return the length of the shortest clear path from top-left to bottom-right (8-directional movement). Return -1 if no path.

Example 1:

Input: `grid=[[0,1],[1,0]]`

Output: 2

Example 2:

Input: `grid=[[0,0,0],[1,1,0],[1,1,0]]`

Output: 4

Example 3:

Input: `grid=[[1,0,0],[1,1,0],[1,1,0]]`

Output: -1

Problem 91

Design Twitter

Tags: Dict | List | Function | Class

Design a simplified Twitter class with: **postTweet(userId, tweetId)**, **getNewsFeed(userId)** (most recent 10 tweets from user and followees), **follow(followerId, followeeId)**, **unfollow(followerId, followeeId)**.

Example 1:

Input: `postTweet(1,5); postTweet(1,3); getNewsFeed(1)`

Output: [3,5]

Example 2:

Input: `postTweet(1,5); follow(2,1); getNewsFeed(2)`

Output: [5]

Example 3:

Input: `postTweet(1,5); follow(2,1); unfollow(2,1); getNewsFeed(2)`

Output: []

Problem 92

Jump Game II

Tags: List | Loop | Function

Given an integer list **nums** where **nums[i]** is the max jump length from index i, return the minimum number of jumps to reach the last index.

Example 1:

Input: nums=[2,3,1,1,4]

Output: 2

Explanation: 0->1->4

Example 2:

Input: nums=[2,3,0,1,4]

Output: 2

Example 3:

Input: nums=[1,2,3]

Output: 2

Problem 93

Sudoku Solver

Tags: List | Recursion | Function | Condition

Write a function that solves a Sudoku puzzle by filling the empty cells ('.') in-place. The board is a 9x9 list of lists.

Example 1:

Input: board=[['5','3','.','.','7','.','.','.','.'], ['6','.','.','1','9','5','.','.'], ['.','9','8','.','.','.','6','.'], ['8','.','.','6','.','.','.','3'], ['4','.','.','8','.','3','.','.','1'], ['7','.','.','2','.','.','.','6'], ['.','6','.','.','.','2','8','.'], ['.','.', '.', '4', '1', '9', '.', '.', '5'], ['.', '.', '.', '8', '.', '.', '7', '9']]

Output: Board solved in-place

Example 2:

Input: board (fully filled)

Output: No change needed

Example 3:

Input: board (single empty cell)

Output: Fill the one empty cell

Problem 94

Text Justification

Tags: String | List | Loop | Function

Given a list of words and a **maxWidth**, format the text so each line has exactly **maxWidth** characters and is fully justified. The last line should be left-justified.

Example 1:

```
Input: words=["This","is","an","example","of","text","justification."],  
maxWidth=16
```

```
Output: ["This is an","example of text","justification. "]
```

Example 2:

```
Input: words=["What","must","be","acknowledgment","shall","be"], maxWidth=16
```

```
Output: ["What must be","acknowledgment ","shall be "]
```

Example 3:

```
Input: words=["a"], maxWidth=5
```

```
Output: ["a "]
```

Problem 95

Flatten and Sort Nested Structure

Tags: List | Dict | Tuple | Recursion | Function

Given a deeply nested structure containing lists, dicts, tuples, and scalar values, return a sorted flat list of all scalar values (numbers and strings).

Example 1:

```
Input: data={'a':[1,(2,3)],'b':{'c':4}}
```

```
Output: [1, 2, 3, 4]
```

Example 2:

```
Input: data=[1,[2,(3,[4])]]
```

```
Output: [1, 2, 3, 4]
```

Example 3:

```
Input: data=(1,'z','a',2)
```

```
Output: [1, 2, 'a', 'z']
```

Problem 96

K Closest Points to Origin

Tags: List | Tuple | Function

Given a list of **points** as (x, y) tuples and integer **k**, return the **k** closest points to the origin (0,0) using Euclidean distance.

Example 1:

Input: points=[(1,3),(-2,2)], k=1

Output: [(-2,2)]

Example 2:

Input: points=[(3,3),(5,-1),(-2,4)], k=2

Output: [(3,3),(-2,4)]

Example 3:

Input: points=[(0,1),(1,0)], k=2

Output: [(0,1),(1,0)]

Problem 97

Task Scheduler

Tags: List | Dict | Loop | Function

Given a list of tasks **tasks** (letters A-Z) and integer **n** (cooldown), return the minimum number of intervals needed to finish all tasks.

Example 1:

Input: tasks=['A','A','A','B','B','B'], n=2

Output: 8

Example 2:

Input: tasks=['A','A','A','B','B','B'], n=0

Output: 6

Example 3:

Input: tasks=['A','A','A','A','A','A','B','C','D','E','F','G'], n=2

Output: 16

Problem 98

Word Break

Tags: String | Dict | List | Function

Given a string **s** and a list of words **wordDict**, return **True** if **s** can be segmented into a space-separated sequence of words from **wordDict**.

Example 1:

```
Input: s="leetcode", wordDict=["leet","code"]
```

```
Output: True
```

Example 2:

```
Input: s="applepenapple", wordDict=["apple","pen"]
```

```
Output: True
```

Example 3:

```
Input: s="catsanddog", wordDict=["cats","dog","sand","and","cat"]
```

```
Output: False
```

Problem 99

Frequency of Most Common Word

Tags: Dict | String | Function | Condition

Given a string **paragraph** and list of banned words **banned**, return the most common non-banned word. Case-insensitive, punctuation is ignored.

Example 1:

```
Input: paragraph="Bob hit a ball, the hit BALL flew far after it was hit.",  
banned=["hit"]
```

```
Output: 'ball'
```

Example 2:

```
Input: paragraph="a.", banned=[]
```

```
Output: 'a'
```

Example 3:

```
Input: paragraph="bob. bob, BOB! bob", banned=[]
```

```
Output: 'bob'
```

Problem 100

Design In-Memory File System

Tags: Dict | List | Class | Function

Design an in-memory file system class with: **ls(path)** (list files/dirs), **mkdir(path)** (create dir), **addContentToFile(filePath, content)**, **readContentFromFile(filePath)**.

Example 1:

```
Input: mkdir('/a/b/c'); addContentToFile('/a/b/c/d', 'hello'); ls('/')
```

```
Output: ['a']
```

Example 2:

```
Input: addContentToFile('/a/b/c/d', 'hello'); readContentFromFile('/a/b/c/d')
```

```
Output: 'hello'
```

Example 3:

```
Input: ls('/a/b')
```

```
Output: ['c']
```